

Solution Stoichiometry

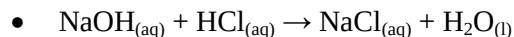
Since solution concentrations are mole based; that is, $n = MV$, this formula can be used to solve stoichiometric calculations involving solutions in chemical reactions.

E.g. 400 mL of a 0.800 mol/L HCl solution is poured over excess zinc metal. Calculate the mass of hydrogen gas produced.

- $\text{Zn}_{(s)} + 2 \text{HCl}_{(aq)} \rightarrow \text{ZnCl}_{2(aq)} + \text{H}_{2(g)}$
- $n_{\text{HCl}} = MV = (0.800 \text{ mol/L})(0.400 \text{ L}) = 0.320 \text{ mol}$
- 2 moles of HCl produces 1 mole of H_2
- 0.320 mol HCl produces x moles of H_2
- $x = \frac{(0.320 \text{ mol})(1 \text{ mole})}{2 \text{ mol}} = 0.160 \text{ mol}$
- $m = n(\text{mm}) = (0.160 \text{ mol})(2.0 \text{ g/mol}) = 0.32 \text{ g}$

When acids and bases are added together, they neutralize each other properties and water and a salt is produced. This involves solution stoichiometry since two solutions are being reacted.

E.g. 35 mL of 0.25 M NaOH solution is used to neutralize 20 mL of an HCl solution. Calculate the concentration of the acid.



$$n_{\text{NaOH}} = M_{\text{NaOH}} V_{\text{NaOH}} = (0.035 \text{ L})(0.25 \text{ mol/L}) = 8.75 \times 10^{-3}$$

1 mole of NaOH reacts with 1 mole of HCl so by stoich, $n_{\text{HCl}} = 8.75 \times 10^{-3}$

$$M_{\text{HCl}} = n/V_{\text{HCl}} = 8.75 \times 10^{-3} / 0.020 = 0.4375 \text{ M}$$

There is a shortcut that can be used because at neutralization, the moles of H^+ added from the acid will exactly equal the moles of OH^- added from base; this is called the point of **equivalence**. But, the H^+ released by the acid is simply due to the number of acidic Hs in the acid chemical formula.

So, the moles of H^+ = (number of acidic H^+)(moles of that acid) and moles of acid = $M_A V_A$.

Likewise for the base; moles of OH^- = (# of OH^- released)(moles of base). At equivalence, moles of H^+ equals the moles of OH^- , so the neutralization equation becomes:

E.g. A 100 mL of 1.5 M HCl is required to neutralize a sample of 1.4 M $\text{Ca}(\text{OH})_2$ solution. Calculate the volume of base.

- $\#_{\text{H}^+} M_A V_A = \#_{\text{OH}^-} M_B V_B$ becomes
- $1(1.5)(0.100) = 2(1.4)V_B$
- $V_B = 1(1.5)(0.100)/[2(1.4)] = 0.05357 \text{ L}$ or 53.57 mL